Diversity Without Difference: Why Architectural Variation Fails to Produce Fusable Speech Deepfake Detectors, and Why the Development Set Cannot Tell You

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Abstract

Fusing several countermeasures is the standard route to strong speech deepfake detection, and the winning ASVspoof 5 system fuses twelve. We ask what the members of such a fusion must actually differ in, and find that architectural variation is not enough: four detectors built on one self-supervised backbone but differing in pooling, representation, topology and even backbone behave as *1.10 independent systems* out of four, with the leading principal component carrying 95.5% of the score variance. We build each detector identically apart from its back-end, measure the eigen-spectrum of their score correlation, and evaluate on the ASVspoof 5 evaluation set and on an out-of-domain in-the-wild set. Only a change of *view*—replacing time-mean pooling with attention over frames—moved the correlation at all (0.994 → 0.917), and our selected two-member system reaches **3.21%** EER / **0.0905** minDCF, ahead of the other published open-condition system we could verify (4.04%) and short of the 2.59% twelve-system winner. Two further results cut against common practice: the ASVspoof 5 development split cannot rank these systems (Spearman $\rho = +0.01$, rising only to +0.24 when a neural-codec view is added), so our protocol selects a system 0.28 EER points worse than the best in our own pool; and the best in-domain member is the *worst* out of domain, with every system degrading by a factor of 2.5–3.8 on found audio.

1 Introduction

Speech deepfake detection is evaluated almost entirely in domain. A countermeasure is trained on one corpus, selected on that corpus's development split, and reported on its evaluation set; the leaderboard orders systems by a single pooled equal error rate. Fusion of several countermeasures is the standard way to the top of such a leaderboard, and the winning ASVspoof 5 open-condition system fuses twelve single systems across three back-end families [Wang et al., 2024b].

This paper asks a narrower question than "how do we win": *what must fusion members differ in for fusion to pay?* We build four detectors that share a training corpus, a frozen self-supervised backbone, an adaptation scheme, an augmentation policy, a seed and a schedule, and differ only in their back-end—time-mean pooling, attentive pooling, a hand-crafted cepstral branch, and graph attention—plus a fifth varying the backbone itself. We then measure not just whether they fuse, but *how many independent systems they amount to.*

The answer is uncomfortable. Across three architectural axes and two backbones, the four members behave as **1.10 effective systems out of four**. Their first principal component explains 95.5% of the score variance and loads within 0.006 of 0.50 on every member: they are, in information terms, one system wearing four hats. The only intervention that moved the

number was a change of *view*—opening the time axis that time-mean pooling discards—and the 0.10 EER points it bought came from 3.2% of the variance.

We report three further findings, each of which complicates the leaderboard reading of this task.

1. **The development split cannot select these systems.** ASVspoof 5's train and development partitions are 100% uncoded while 75% of the evaluation set is coded. Over all 15 member subsets, the rank correlation between development and evaluation EER is $\rho = +0.011$ ($p = 0.97$). Adding a neural-codec development view raises it only to $+0.236$ ($p = 0.40$). Our protocol therefore selects a system 0.28 EER points worse than the best in our pool, and we report both numbers.
2. **In-domain rank inverts out of domain.** On an in-the-wild set, our best in-domain member is the worst, and our third-best is the best by a wide margin. All systems degrade by 2.5–3.8×.
3. **The evaluation set is smaller than it looks.** Its 680,774 clips derive from 171,602 source utterances, each appearing clean plus roughly three codec variants. Bootstrap intervals that resample clips—ours in earlier drafts, and, we suspect, most reported on this corpus—are about 2× too narrow.

2 Related work

Self-supervised speech representations are the dominant front end for this task. We use WavLM-Large [Chen et al., 2022]; XLS-R [Babu et al., 2021] is our second backbone. Low-rank adaptation [Hu et al., 2022] lets us fine-tune a 300M-parameter backbone with 3.15M trainable parameters.

On back-ends, AASIST [Jung et al., 2022] introduced heterogeneous spectro-temporal graph attention and is the architecture family our graph member is derived from; we reuse its graph attention layer but not its heterogeneous cross-attention or master node, and we are explicit that our variant is not AASIST. Attentive pooling over frozen SSL layers, and automatic layer selection, are studied by Serrano et al. [2025], who also report that score-level fusion of several encoders improves out-of-domain generalisation.

For augmentation we use RawBoost [Tak et al., 2022] and simulated codecs, and we add the neural codec EnCodec [Défossez et al., 2022] during training. The adversarial attacks present in the ASVspoof 5 evaluation set are Malafide [Panariello et al., 2023] and Malacopula [Todisco et al., 2024].

Our comparison points on ASVspoof 5 track 1 are the challenge overview [Wang et al., 2024a], the winning USTC-KXDIGIT system at 2.59% EER / 0.0750 minDCF in the open condition [Wang et al., 2024b], and SZU-AFS at 4.04% / 0.115 [Xu et al., 2024]. Both report on the full evaluation set under the official cost model, as we do.

3 Data

Table 1 and Figure 1 describe the corpora. Three properties of ASVspoof 5 [Wang et al., 2024a] drive everything that follows.

Disjoint attacks. A01–A08, A09–A16 and A17–A32 do not overlap, so a system is always tested on spoofing algorithms it has never seen. This design repeatedly caught training-attack memorisation in our own runs that training loss hid completely.

A channel mismatch between selection and test. Train and development are uncoded. The evaluation set is 75% coded, with eleven conditions. We simulate ten of the eleven during training; only C11 (device variation) is never simulated. Section 6.4 shows that this mismatch, and not any property of the models, is what breaks model selection.

Table 1: Corpora. ASVspoof 5 attack sets are disjoint across partitions, so every reported number is on unseen spoofing systems. Train and development are entirely uncoded; 75% of the evaluation set carries one of eleven real codecs.

Partition	Clips	Bona fide	Speakers	Attacks	Channel
ASVspoof 5 train	182,357	18,797 (10.3%)	400	A01–A08	uncoded
ASVspoof 5 dev	140,950	31,334 (22.2%)	785	A09–A16	uncoded
ASVspoof 5 eval	680,774	138,688 (20.4%)	737	A17–A32	clean + C01–C11
In-the-Wild	31,779	19,963 (62.8%)	54	unlabelled	unknown

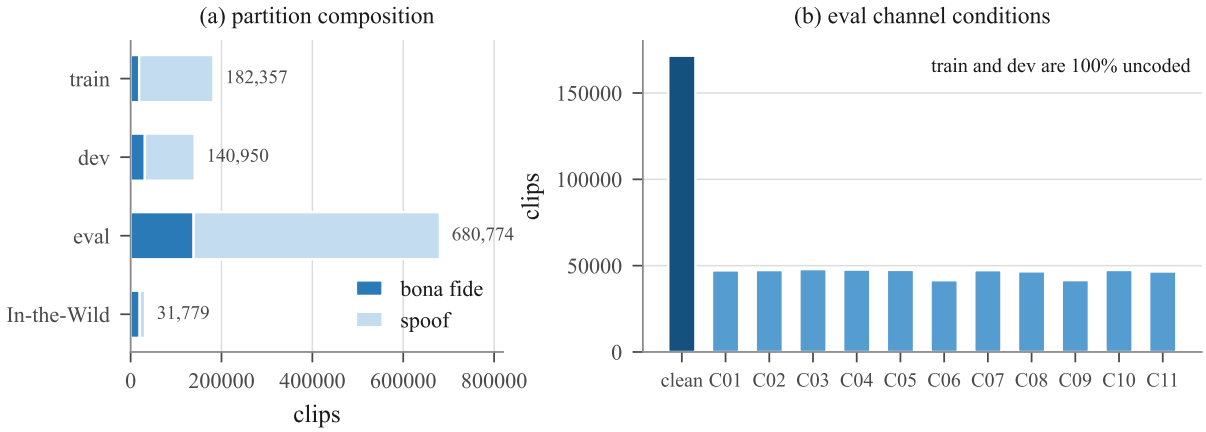


Figure 1: (a) Partition composition. (b) Evaluation-set channel conditions. The selection split contains none of the eleven coded conditions the evaluation set is judged on.

A source-utterance structure that inflates the apparent sample size. Every coded evaluation clip carries a `CODEC_SEED` pointing at its original, and all 509,172 coded clips have their original also present. The 680,774 clips therefore come from **171,602 source utterances**, a mean of 3.97 clips per source. Treating clips as independent in a bootstrap understates every interval by roughly $\sqrt{4}$. Intervals in this paper are computed with a clustered bootstrap over source utterances.

In-the-Wild is used for reporting only; nothing is selected or calibrated on it. Note its inverted class prior (62.8% bona fide), which makes minDCF under the official 0.05 spoof prior incomparable with the ASVspoof numbers; we report EER there.

4 Systems

Figure 2 shows the common pipeline. A waveform is augmented, then passed through WavLM-Large [Chen et al., 2022] adapted with LoRA of rank 16 on the attention projections [Hu et al., 2022] (3.15M trainable parameters). The backbone emits 25 layer activations. Augmentation applies one of eight simulated codecs with probability 0.5 and, with probability 0.3, EnCodec [Défossez et al., 2022] on the GPU—the neural codec corresponding to evaluation conditions C04 and C07, which the DataLoader cannot apply because it requires a GPU.

The four back-ends are:

Pool (0.26M parameters) takes the time-*mean* of each layer, learns a softmax weighting over layers, and applies a two-layer MLP. This is the conventional design and the baseline for everything else.

MHFA (0.53M) keeps the frames. It learns two *separate* layer weightings—one producing keys, one producing values—and pools over time by multi-head attention, so the layers that decide *where* to look need not be the layers carrying the evidence.

LFCC (0.50M) adds a second *representation*: 20 linear-frequency cepstral coefficients with

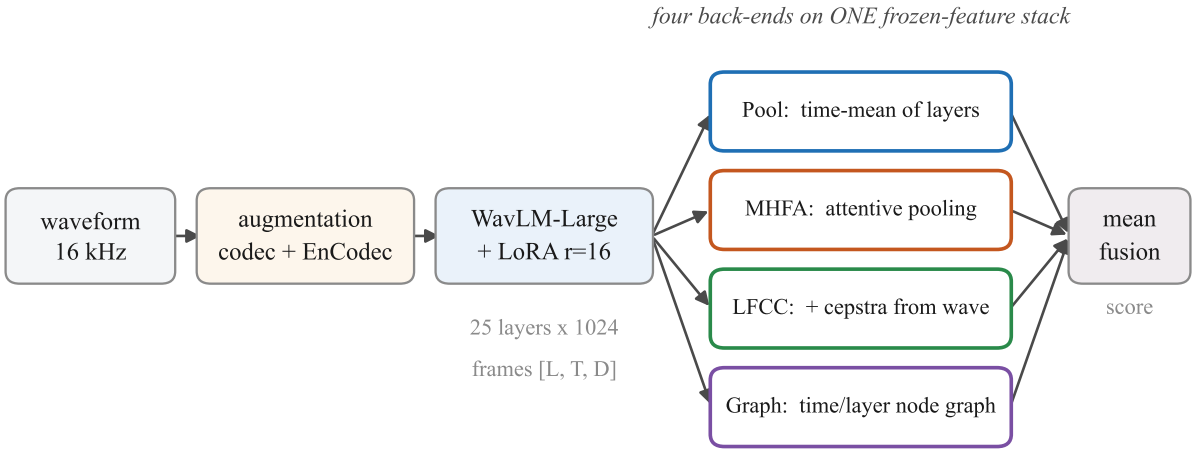


Figure 2: All four members share the waveform pipeline, the augmentation policy and the adapted WavLM-Large backbone, and differ only in the back-end that turns per-layer frames into a score.

first and second deltas, computed from the waveform and aligned frame-for-frame with the backbone, concatenated with the frames before attentive pooling. Part of its evidence cannot correlate with anything the backbone encodes.

Graph (0.30M) adds a second *topology*: graph attention over two node sets, time slots and backbone layers, reusing the graph attention layer of AASIST [Jung et al., 2022]. It is not AASIST— there is no heterogeneous cross-attention and no master node, and the spectral sub-graph is replaced by a layer sub-graph because SSL frames have no frequency axis—and no result here should be read under that name.

Members are combined by the unweighted mean of scores standardised on the development `calib` split. The combiner has no parameters fit on evaluation data.

5 Protocol

Models are selected on the development `select` split (attacks A09–A16), scale statistics are fit on the speaker-disjoint `calib` half, and the evaluation set is scored once. Metrics are EER and minDCF under the official ASVspoof 5 cost model (spoof prior 0.05, $C_{\text{miss}} = 1$, $C_{\text{fa}} = 10$), plus $\min C_{\text{llr}}$. Differences are tested with a paired bootstrap clustered on source utterance (Section 3).

All members use seed 1337, batch size 32, eight epochs, head learning rate 10^{-3} and backbone learning rate 10^{-5}. **A limitation we state up front:** the members below are single-seed. In earlier three-seed experiments on this codebase, frozen single models varied by 0.33 EER points across seeds—larger than the 0.10-point fusion gain we report. Seed variance for the systems in Table 2 is therefore *unmeasured*, and the fusion gain should be read with that in mind.

6 Results

6.1 Members

Table 2 is the central table of the paper, and almost every column disagrees with every other. Pool is the best member on development and only third on evaluation. LFCC is the *worst* on development, the *best* on evaluation, and the *worst* again in the wild. MHFA is third in domain and first out of domain by 2.7 points.

Figure 3 shows that these differences are not training artefacts: none of the members shows the rising-development / falling-training signature of attack memorisation, and the LFCC member simply peaks at epoch 1 and then slowly overfits. Figure 4 gives the detection error

Table 2: The four members. All share everything but the back-end. Dev select is the mean EER over the clean and codec development views; in-the-wild is out-of-domain and reporting only. ρ_{Pool} is the Pearson correlation of evaluation scores with the Pool member.

			ASVspoof 5 eval			In-the-Wild	
Member	Back-end differs in	Dev sel.	EER	MINDCF	minC_{llr}	EER	ρ_{Pool}
Pool	—	**1.50**	3.31	0.0942	0.1256	11.25	—
MHFA	pooling	2.07	3.44	0.0957	0.1286	**8.51**	0.917
LFCC	representation	2.95	**3.11**	**0.0845**	**0.1164**	11.82	0.965
Graph	topology	3.10	3.93	0.1091	0.1475	11.44	0.943

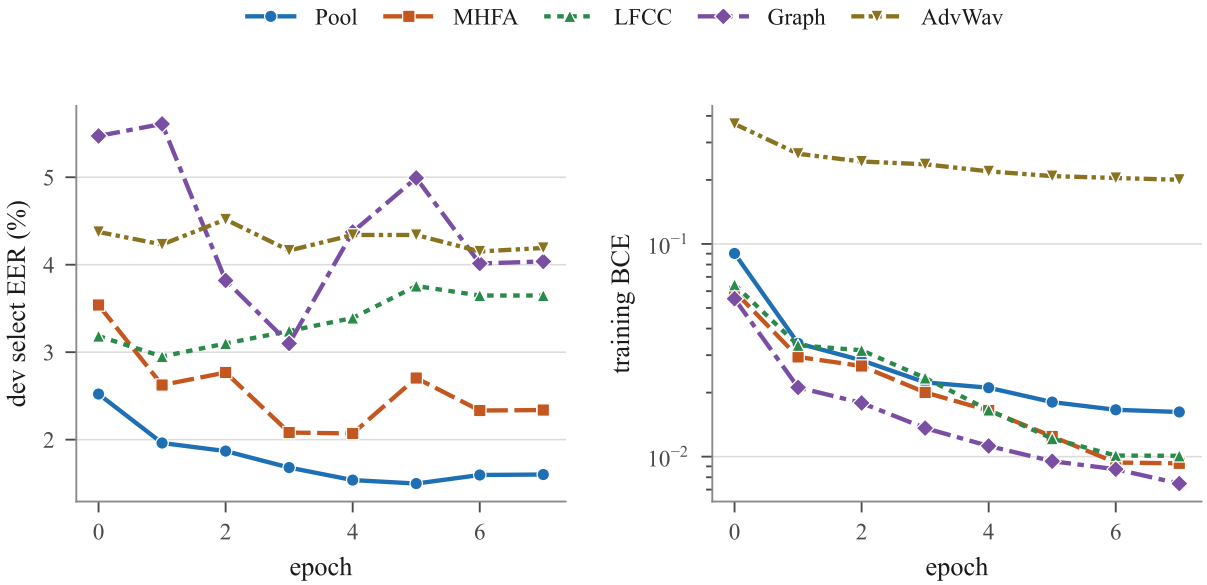


Figure 3: Training behaviour. Left: development EER. Right: training loss. AdvWav is the adversarially trained model of Section 6.6; it converges to roughly double the development EER of every other member.

trade-off curves; the fused systems dominate the single members across the operating range, not only at the equal error point.

6.2 Conditions and attacks

Figure 5 breaks the evaluation set down by channel. The two neural codecs, C04 and C07, are by a wide margin the hardest conditions for every member—a direct consequence of their absence from any uncoded training view before we added GPU-side EnCodec. The right panel of Figure 4 breaks it down by attack: **A28 alone is 9.36%** while nine of sixteen attacks are under 1.5%, so the pooled EER is largely a statement about one attack. The adversarial attacks [Panariello et al., 2023, Todisco et al., 2024] are *not* the difficult ones.

The attacker-adaptation conditions are flat—AC1 3.29%, AC2 3.14%, AC3 3.19%—so how hard the attacker adapted against a surrogate detector made no measurable difference to us.

6.3 How many systems is a four-member ensemble?

We summarise a member set by the eigen-spectrum of its score correlation matrix, reporting the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ and the entropy rank $\exp(-\sum_i p_i \log p_i)$ with $p_i = \lambda_i / \sum_j \lambda_j$. Both equal 1 when all members are the same system and M when they are orthogonal.

Figure 6 gives the answer: **1.10 effective systems out of four** (entropy rank 1.26), with PC1 explaining 95.5% of the variance and loading $0.501, 0.496, 0.501, 0.502$. On the out-of-domain

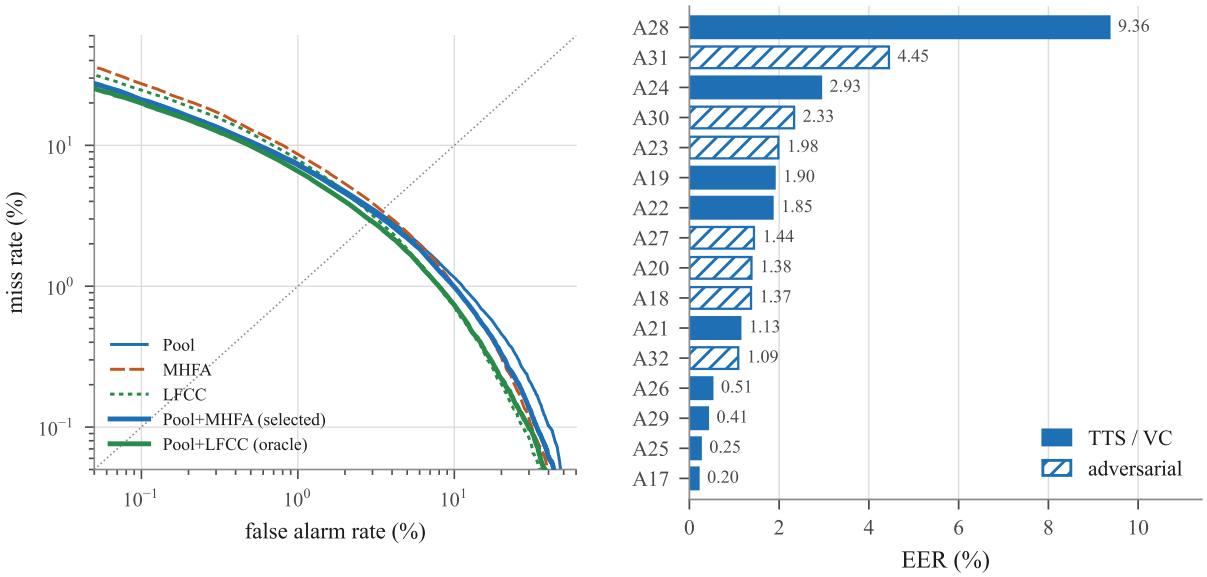


Figure 4: Left: DET curves on the evaluation set; solid curves are fusions. Right: per-attack EER of the selected system. The pooled number is dominated by A28; nine of sixteen attacks sit below 1.5%, and the adversarial attacks (hatched) are not the hard ones.

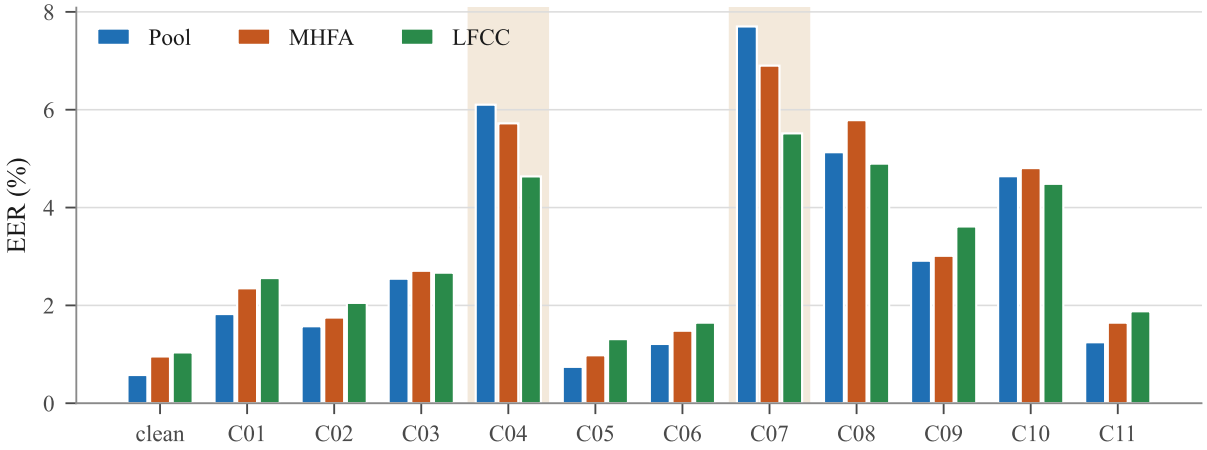


Figure 5: Per-condition EER. The neural codecs C04 and C07 (shaded) are the hardest conditions for every member, and the development split contains neither.

set the figure is 1.15 of 4. Changing the pooling, the representation, the topology *and* the backbone moves the effective rank by 0.1.

This reframes the fusion question. Table 3 shows the gain available: the best two-member fusion improves on its best member by 0.18 EER points, and a third member adds nothing. The relevant comparison is Wang et al. [2024b], whose twelve systems across three back-end families gain 35% relative from fusion. Our members do not differ enough for that mechanism to engage, and the axis that moved them at all was the one that changed what the model *sees*, not how it is parameterised: replacing time-mean pooling with attention over frames took the correlation from 0.994 (for a member differing only in augmentation strength) to 0.917.

6.4 Selection fails, and we report both numbers

Over all 15 subsets the rank correlation between development and evaluation EER is $\rho = +0.011$ ($p = 0.97$): the development split carries *no* information about evaluation ordering (Figure 7a). Since the evaluation set's discriminating conditions are real codecs and the development split has none, we scored every member through an added neural-codec development view. This

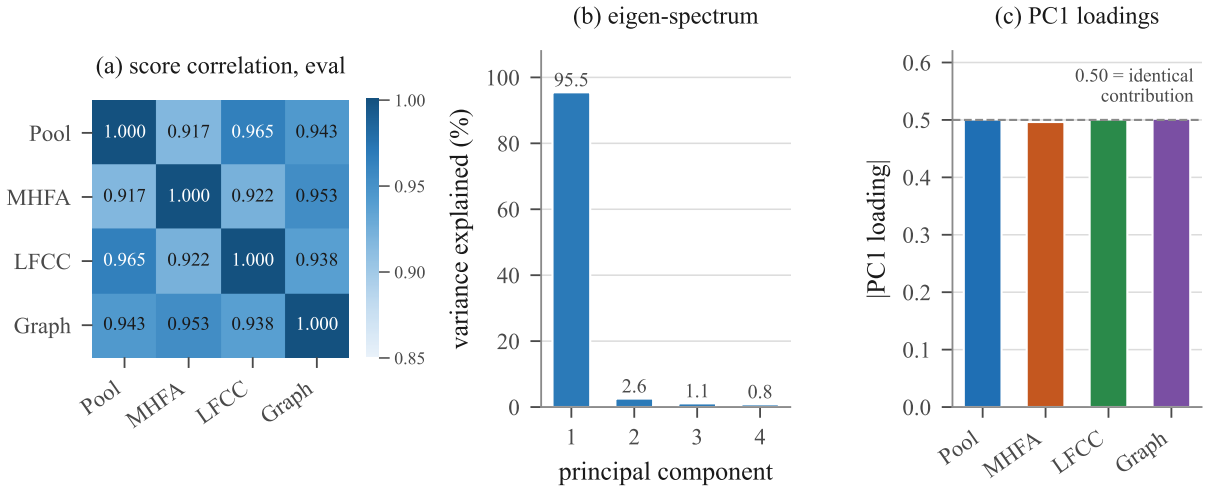


Figure 6: (a) Score correlation on the evaluation set. (b) Eigen-spectrum of that matrix: the first component explains 95.5% of the variance. (c) Its loadings are within 0.006 of 0.50 on every member—the first component *is* the unweighted mean.

Table 3: Member subsets ranked by development EER, top four of fifteen, plus the best subset on evaluation. The protocol selects row 1. The oracle row is shown for comparison and is *not* claimed as a result.

Rank	Subset	Dev sel.	eval EER	eval minDCF	ITW EER
Selection on clean + codec ($\rho = +0.011$, $p = 0.97$)					
1	Pool + MHFA	**1.46**	3.21	0.0905	9.04
2	Pool	1.47	3.31	0.0942	11.25
3	Pool + MHFA + Graph	1.67	3.20	0.0903	8.80
4	Pool + MHFA + LFCC	1.72	2.94	0.0827	9.04
7	*Pool + LFCC* (oracle)	1.83	*2.93*	*0.0819*	9.19
Selection with a neural-codec view added ($\rho = +0.236$, $p = 0.40$)					
1	Pool + MHFA	**2.79**	3.21	0.0905	9.04
2	Pool + MHFA + LFCC	2.90	2.94	0.0827	9.04
6	*Pool + LFCC* (oracle)	3.01	*2.93*	*0.0819*	9.19

helps—the three-member system climbs from rank 4 to rank 2, and ρ rises to +0.236—but at $p = 0.40$ it is still not distinguishable from zero (Figure 7b).

The consequence is Table 3. Our protocol selects Pool + MHFA at **3.21% / 0.0905**. The best system in our pool is Pool + LFCC at *2.93% / 0.0819*, which the protocol ranks seventh of fifteen. **We report the first as our result and the second as an oracle bound, and we take the 0.28-point gap between them to be a measurement of the selection set rather than a property of our models.** We are not aware of a prior quantification of this gap on ASVspoof 5.

6.5 In-domain and out-of-domain rankings invert

Figure 8 (left) plots each member in domain and out. Every system loses a factor of 2.5–3.8, which is in line with the literature. What is not routine is the inversion: LFCC, our best in-domain member at 3.11%, is our worst in the wild at 11.82%, while MHFA goes from third (3.44%) to first (8.51%), a 24% relative margin over the next system.

A plausible reading is that cepstral evidence is partly *channel* evidence, and channel statistics are exactly what does not transfer from a single curated corpus to found audio, whereas attentive pooling over time learns something less corpus-specific. We offer this as a hypothesis; our experiments establish the inversion, not its cause.

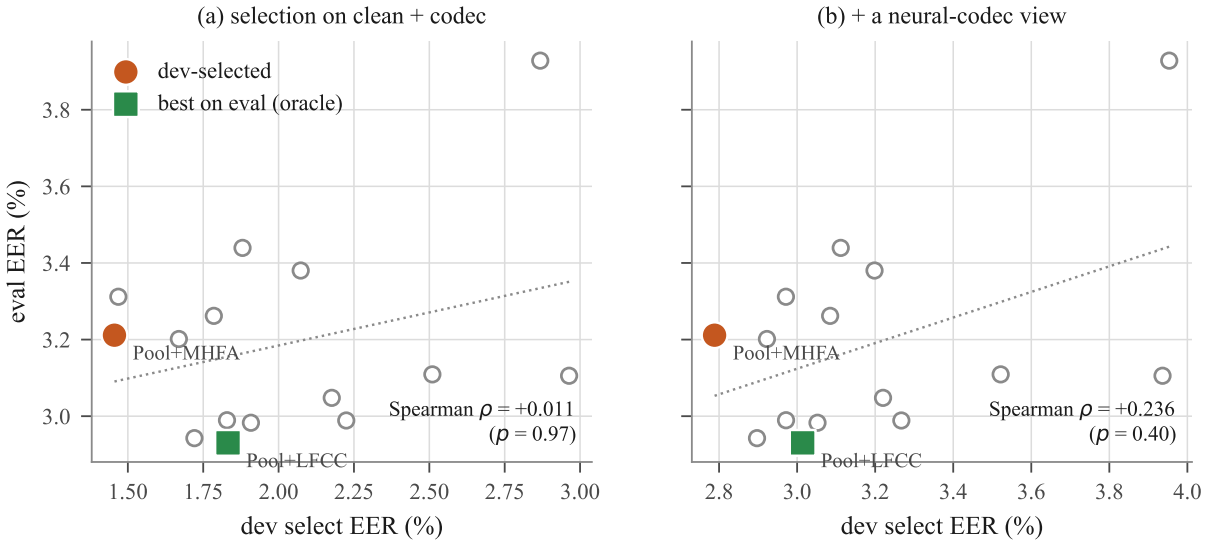


Figure 7: Development EER against evaluation EER for all 15 member subsets. The dotted line is a least-squares fit. Adding a neural-codec view to the selection set improves the ordering—the three-member system climbs from rank 4 to rank 2—but does not make it predictive.

The practical implication is uncomfortable for leaderboard optimisation. Pursuing the final 0.6 EER points in domain would mean preferring exactly the member that transfers worst.

6.6 Adversarial robustness

Table 4 summarises the robustness measurements. A gradient-bearing attacker succeeds on essentially every clip at perturbations of 30 dB SNR against four of the five defences: a single model, a four-member band ensemble, each of two SSL backbones used as the other's surrogate, and after feature-space adversarial training—which additionally destroyed clean accuracy (13.95% EER). A query-limited black-box attacker with 320 queries per clip succeeds on about 1%.

Waveform adversarial training is the one route that moves. Hardening the path the attacker actually takes—half of each batch replaced by 30 dB PGD examples crafted through the backbone—drops white-box success from 1.000 to 0.953 at 30 dB, and transfer success from 1.000 to 0.739. Those are the first sub-unity white-box numbers in this table. Its own control, trained identically without the adversarial half, sits at 1.000 on every row, so the drop is attributable to the defence rather than to the schedule.

The price is severe and we state it plainly. Clean evaluation EER rises from 3.31% to 7.09%, a paired difference of +0.0378 with a 95% interval of [+0.0372, +0.0384] over all 680,774 clips, and the damage concentrates on the codec conditions (C07 11.06%, C08 10.08%). Buying a 4.7-point reduction in white-box attack success for 3.78 EER points of clean accuracy is not a trade a deployed system would take. Our contribution here is therefore a measurement rather than a method: under this threat model the SSL detectors in this paper are not defensible at usable accuracy, and the one defence that attacks the right surface shifts that frontier without crossing it.

We flag one methodological point about the feature-space row. That defence hardened the *head* against perturbations in *feature* space, while the attacker perturbs the *waveform* and arrives through the backbone. The mismatch means its failure was weak evidence, and the waveform run above is the corrected version of that experiment. The correction mattered in both directions: it recovered 6.9 EER points of clean accuracy relative to the feature-space attempt (7.09% against 13.95%), and it was the change that moved white-box success at all.

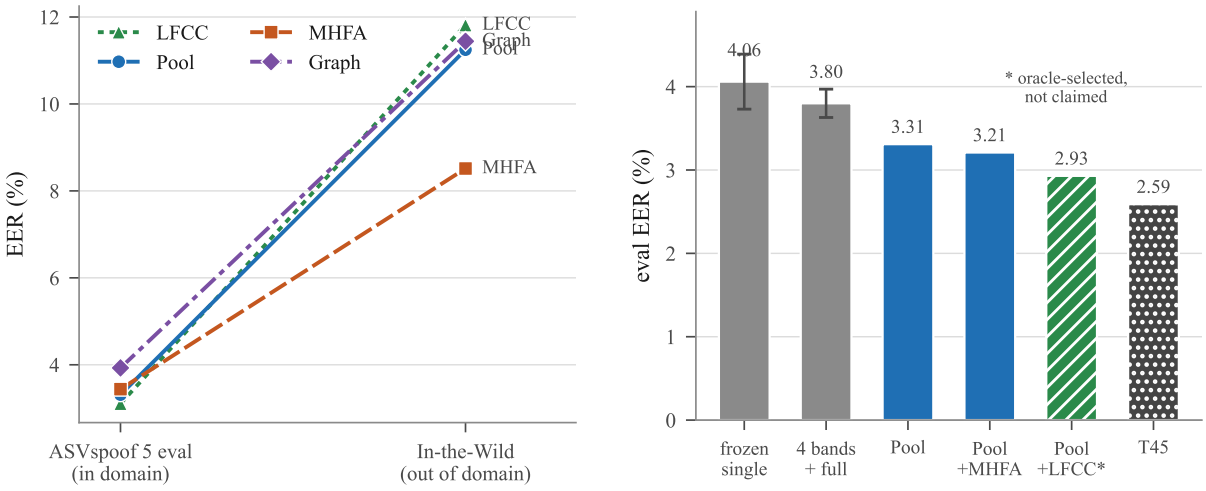


Figure 8: Left: every system degrades by 2.5–3.8× out of domain, and the ordering inverts. Right: progression of this work. Bars with error bars are three-seed means with spread; the remainder are single-seed. The hatched bar is oracle-selected and not claimed.

7 Limitations

We state these rather than leave them to be found.

Single seed. Every system in Table 2 is one seed. Three-seed experiments earlier in this work gave a spread of 0.33 EER points on frozen single models, larger than the 0.10-point fusion gain. The fusion gain is therefore not established against seed variance.

One corpus. We train on ASVspoof 5 alone. The system we compare against [Wang et al., 2024b] trains on four corpora. We cannot separate "our architecture is worse" from "our supervision is narrower".

Clustered intervals matter. Section 3 shows that clip-level bootstrap intervals on this corpus are about 2× too narrow. Our conclusions survive the correction because the differences we claim are many standard errors wide, but intervals reported elsewhere on this corpus may not.

One out-of-domain set. In-the-Wild has its own biases—public figures, internet audio, an inverted class prior. One such set measures a gap; it does not characterise generalisation.

The graph member is not AASIST and must not be cited as evidence about AASIST.

8 Conclusion

Four detectors differing in pooling, representation, topology and backbone amount to 1.10 independent systems. Architectural variety is not the same as error diversity, and only a change in what the model *sees* moved the correlation at all. Meanwhile the development split of the corpus everyone uses cannot rank these systems, costing our protocol 0.28 EER points against its own oracle; and in-domain ranking inverts out of domain, so the member that wins the leaderboard is the one that transfers worst.

Our selected system reaches 3.21% EER / 0.0905 minDCF, ahead of the other published open-condition system we were able to verify [Xu et al., 2024, 4.04% / 0.115] and short of the twelve-system winner at 2.59% / 0.0750 [Wang et al., 2024b]. A further comparison point of 4.17% / 0.119 for the best single system appears in our own experiment log, but we could not verify its source at the time of writing and therefore do not rely on it [CITATION NEEDED]. We think the more useful contribution is the measurement around that number: a reproducible way to ask how many systems an ensemble really contains, and evidence that the standard selection procedure on this benchmark does not work.

Table 4: Attack success at a 0.1% false-alarm operating point, 20-step PGD under an SNR-constrained L_2 budget; realised SNR is reported, not assumed. Generated directly from the attack logs, so the table cannot drift from the measurements. Four routes are closed outright; waveform adversarial training is the only one that moves white-box success off 1.000.

Defence	Attack	Clean EER	30 dB	20 dB
single model	white-box PGD	4.78	1.000	1.000
single model	SPSA	4.78	—	0.011
four layer bands	white-box PGD	4.78	1.000	1.000
four layer bands	transfer PGD	4.78	1.000	1.000
four layer bands	SPSA	4.78	—	0.014
WavLM, surrogate XLS-R	white-box PGD	4.78	1.000	1.000
WavLM, surrogate XLS-R	transfer PGD	4.78	0.892	0.995
XLS-R, surrogate WavLM	white-box PGD	9.21	1.000	1.000
XLS-R, surrogate WavLM	transfer PGD	9.21	0.832	0.984
feature-space adv. training	white-box PGD	13.95	0.993	1.000
feature-space adv. training	transfer PGD	13.95	0.776	0.977
waveform adv. training	white-box PGD	7.07	0.953	0.967
waveform adv. training	transfer PGD	7.07	0.739	0.951
its control (Pool)	white-box PGD	4.17	1.000	1.000
its control (Pool)	transfer PGD	4.17	1.000	1.000

Reproducibility

Figures are generated by `paper/make_figures.py` from a single JSON of measured values, so no number in the text can disagree with a figure. Every experiment, including the failures and the predictions that turned out wrong, is recorded in `docs/experiment_log.md`.

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